

Werner Bisschoff

werner@bisschoff.dev
071 826 2066
Cape Town, Western Cape, South Africa

Enterprise Systems & Automation Architect

Professional Summary

Strategic Enterprise Software Architect with 5+ years of experience engineering robust business automation platforms, custom ERP extensions, and scalable API layers. Expert in the Frappe/ERPNext ecosystem, asynchronous workflow systems, and cross-platform mobile application deployment (Expo/React Native). Proven track record of translating complex real-world operations into scalable, type-safe data schemas governed by strict Spec-Driven Development (SDD) paradigms.

Experience

Founder and Host Cape Town	Divergent Tabletop Jul 2025
• Built multi-tenant isolation using PostgreSQL Row-Level Security (RLS) and SET LOCAL transaction-scoped context injection. • Used Ecto.Multi atomic chains and Oban transactional background workers for multi-stage backend edits. • Built real-time collaborative editing via Phoenix LiveView + WebSockets + Phoenix.PubSub. • Implemented per-page EditLock GenServer with heartbeat timeout and fair handover queue to prevent concurrent write conflicts. • Applied Monotropic focus paradigms and Universal Design principles to minimize platform cognitive complexity.	
Full-Stack Software Engineer Cape Town	FARO Africa Aug 2024 – Nov 2025
• Extended ERPNext using Python/JavaScript to improve workflows, pricing logic, and operational reporting (SQL) → reduced manual reporting time and improved data accuracy • Migrated internal mobile Retool workflows to Expo → improved performance and enhanced long-term maintainability • Provisioned AWS infra with Pulumi and deployed services including Inngest and PayloadCMS. • Diagnosed and resolved issues in a large existing ERPNext installation. • Introduced LLM-assisted development workflows → improved debugging speed and code review throughput	
Embedded Software Engineer Remote Work	Ingenics Digital GmbH (through ViVa Outsourcing) Mar 2023 – May 2024
• Built Python-based tooling for serial/BLE communication, including client-facing test executables → accelerated testing and debugging workflows • Created Python hardware mocks for rapid iteration and early-stage testing → enabled faster development cycles with fewer hardware dependencies • Managed fast, reliable data interchange using a compact TinyFrame binary protocol	

- Implemented asynchronous communication workflows with Python and Pytest

Junior Lecturer

Potchefstroom

North-West University

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Education

B.Eng. Computer and Electronic Engineering

Potchefstroom

North-West University

2020

- Focus on embedded systems, software engineering, and electronic design
- Developed data analysis pipeline with Python/Pandas (linear regression, ML); built NFC payment emulator (Kotlin/Android)

Skills

Primary: Python, ERPNext/Frappe Framework, Expo (React Native), Inngest

Secondary: AWS, Pulumi (IaC), PostgreSQL (RLS), Docker

Foundational: SQL, Object-Oriented Design, Spec-Driven Development

Other: C++, FreeRTOS (Low-level integration support)

AI Policy

I treat AI as an agentic partner within a strict engineering framework, moving beyond “vibe-coding” to structured, audit-ready workflows:

Spec-Driven Development (SDD): Prioritize context-heavy specifications to ensure alignment; AI-generated outputs are treated as hypotheses validated by human-led TDD.

Data Security: Enforce mandatory sandboxing for all agentic projects; strictly prohibit credential access and utilize tool-based retrieval for scoped, anonymized data only.

Engineering Rigour: All AI-assisted output undergoes rigorous human-in-the-loop review to ensure it meets enterprise quality and compliance benchmarks.

Privacy by Design: AI workflows are architected to avoid direct database access; sensitive data remains behind scoped, tool-mediated retrieval with no standing credentials exposed to agents.

Job Target

Seeking a mid to senior software engineering role (remote preferred, hybrid max. 2 days/week on-site in Cape Town) with emphasis on embedded systems, backend services, or AI-assisted development workflows. Prefers small, focused teams with clear goals and engineering culture. Open to permanent or contract (3+ months) arrangements.